ECO 202 — In-Class Exam 2

Statistics and Data Analysis for Economics | Princeton University

October 12, 2026 | Classes 6–9 | 100 points

Name: _____ Student ID: _____

Timing. Aim to complete your answers in 55–60 minutes, leaving 10–15 minutes to check within the 70-minute target. Follow announced start/end times within the 80-minute meeting.

Conditions. No books, notes, calculators, AI, or other electronics; University-authorized accommodations apply. Fractions, radicals, and equivalent exact expressions are acceptable. Show reasoning in written answers. Data are teaching examples unless identified otherwise.

Points. Questions 1–4: one answer A–D, 5 points each, no negative marking. Questions 5–8: 20 points each; subpart points are shown. Answer each part in its space; scratch pages 3 and 12 are not graded.

Contents and suggested pacing

Question	Topic	Page	Points	Minutes
1	Sampling and nonresponse	2	5	2.5
2	Random assignment and causal interpretation	2	5	2.5
3	Validity of probabilities	2	5	2.5
4	Conditional probability	2	5	2.5
5	Equal inclusion and simple random sampling	4	20	12
6	An experiment and its causal target	6	20	15
7	Events, overlap, and independence	8	20	12
8	Updating the likely source of a shipment	10	20	10
Checking and review			—	11
Total			100	70

Reference formulas and scratch work: page 3.

Princeton University Honor Code

At the end of the examination, write the pledge in full below and sign your name.

"I pledge my honor that I have not violated the Honor Code during this examination."

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Signature: _____

Multiple choice (20 points; about 10 minutes)

1. A university selects an SRS of 200 students from a complete roster, but only 80 reply. Which statement is justified?

A. Nonresponse can change the sample composition if response relates to the outcome.

B. The response rate alone determines the direction of nonresponse bias.

C. The respondents must be an SRS because the initial selection was random.

D. Increasing the original sample always eliminates nonresponse bias.

2. Volunteers are randomly assigned to receive or not receive an invitation to job training. With complete outcome collection and no interference, the comparison by assigned group principally supports:

A. an average causal effect of the invitation for the study participants.

B. the exact effect of completing training for each individual participant.

C. an average causal effect for all unemployed people without further assumptions.

D. proof that the two groups have identical baseline characteristics.

3. Which proposed probabilities are impossible for any events A and B?

A. $\mathbb{P}(A) = 1/2$, $\mathbb{P}(B) = 1/2$, $\mathbb{P}(A \cap B) = 1/4$.

B. $\mathbb{P}(A) = 1/2$, $\mathbb{P}(B) = 1/2$, $\mathbb{P}(A \cap B) = 0$.

C. $\mathbb{P}(A) = 1/2$, $\mathbb{P}(B) = 1/2$, $\mathbb{P}(A \cap B) = 1/2$.

D. $\mathbb{P}(A) = 3/4$, $\mathbb{P}(B) = 3/4$, $\mathbb{P}(A \cap B) = 1/4$.

4. A fraud flag appears on 4/5 of fraudulent transactions. What further information is needed to find the probability a flagged transaction is fraudulent?

A. Nothing: that probability is 4/5.

B. Only the probability a fraudulent transaction receives no flag.

C. Only the total number of transactions.

D. The fraud base rate and the flag rate among nonfraudulent transactions.

Reference formulas and scratch work

Supplied formulas. You may use

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B),$$

$$\mathbb{P}(A) = \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c), \qquad \mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Conditional formulas require positive conditioning probabilities. If asked to derive a rule, give the derivation rather than citing that rule. Any needed counts of unordered subsets are provided in the question.

Scratch work. Use the remaining space for calculations. Work on this page is not graded. Written Questions 5–8 begin on the following pages.

5. Equal inclusion and simple random sampling (20 points; about 12 minutes)

A finite population contains five firms, A, B, C, D, and E. A sampling design selects one of the following five unordered triples, each with probability 1/5:

$$ABC, \quad BCD, \quad CDE, \quad DEA, \quad EAB.$$

There are ten distinct unordered triples of five firms.

(a) (4 points) Explain why this is a probability-sampling design. Find the probability that firm A is selected.

(b) (6 points) Verify that all five firms, A, B, C, D, and E, have the same inclusion probability (probability of being selected). Does this design produce a simple random sample (SRS) of three firms? Justify your answer by giving a triple that an SRS can produce but this design cannot.

(c) (6 points) Suppose the selected triple is ABC. Let y_i denote firm i's number of employees. Write a formula for the population mean of the five firms and a formula for the observed sample mean in terms of y_A, y_B, y_C, y_D, and y_E. Does the equality of inclusion probabilities guarantee that these two means are equal in this realized sample? Justify your answer.

(d) (4 points) Now suppose firms A, B, C, D, and E are the local business council's members, while the target population is all firms in the city. What additional problem arises when using the sample mean from (c) to estimate this population mean? Would including all five council members in the sample solve that problem? Explain.

6. An experiment and its causal target (20 points; about 15 minutes)

Four volunteers are enrolled in an experiment. Exactly two are randomly assigned an offer of tutoring, with all six pairs equally likely. The outcome is the number of exercises solved correctly. The fictional complete potential-outcome table is shown for this question only; in an actual experiment one potential outcome per person would be unobserved.

Person	$Y_i(0)$: no offer	$Y_i(1)$: offer
A	3	5
B	5	6
C	2	5
D	8	10

(a) (4 points) Calculate the individual effects and the average treatment effect for these four volunteers. Include the outcome units.

(b) (6 points) If A and D receive the offer, list all four observed outcomes and calculate the offered-minus-unoffered difference in observed means.

(c) (6 points) Suppose instead that B and C receive the offer.

(i) Calculate the offered-minus-unoffered difference in observed means.

(ii) Take as given that the difference in (i) does not equal the average treatment effect in (a), and that the average no-offer outcome $Y_i(0)$ for B and C differs from that for A and D. Does either fact prove that randomization failed? Explain.

(iii) What does averaging the offered-minus-unoffered difference in observed means over all six assignments recover? No enumeration is required.

(d) (4 points) Suppose some volunteers assigned an offer do not attend tutoring. Consider two comparisons of outcomes: (1) volunteers assigned an offer versus those assigned no offer; (2) volunteers who attended tutoring versus those who did not. Which comparison retains the randomized assignment, and what effect does it estimate? Explain why the other comparison need not identify the causal effect of attending tutoring.

7. Events, overlap, and independence (20 points; about 12 minutes)

Let A and B be events with $\mathbb{P}(A) = 3/5$, $\mathbb{P}(B) = 1/2$, and $\mathbb{P}(A \cap B) = 1/5$.

(a) (8 points) Find the probabilities of at least one event, neither event, and exactly one event. Make the meaning of "exactly one" explicit.

(b) (6 points) Find $\mathbb{P}(B \mid A)$ and $\mathbb{P}(A \mid B)$. Identify each conditioning event and its denominator.

(c) (6 points) Are A and B disjoint? Are they independent? Justify each answer separately.

8. Updating the likely source of a shipment (20 points; about 10 minutes)

A retailer receives shipments from two sources, North and South. Select a shipment at random from the retailer's stated probability model. Let N mean the shipment came from North, and L mean it arrived late. The model gives

$$\mathbb{P}(N) = \frac{2}{5}, \qquad \mathbb{P}(L \mid N^c) = \frac{7}{10}, \qquad \mathbb{P}(L \cap N) = \frac{1}{10}.$$

(a) (8 points) Construct the full two-by-two table of joint probabilities for source (North or South) and arrival (late or not late), including its row and column totals.

(b) (8 points) A shipment is late. Find the probability that it came from North. Explain why $\mathbb{P}(L \mid N)$ is a different probability and calculate that probability too.

(c) (4 points) Are source and lateness independent in this model? Does learning that a shipment is late raise or lower the probability it came from North? Justify without decimal approximations.

Additional scratch work

Use this page for additional calculations. Work on this page is not graded.